Lab 1: Requirements Table

Name:Jamula Soham Rao

SRN:PES2UG24CS202

Problem Statement #15 — Pharmacy Expiry & Re-order Dispatch Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall enforce FEFO (First-Expired-First-Out) picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	High	Pass: Oldest valid batch selected first. Fail: Newer batch assigned while an older unexpired batch exists.	Prevents wastage of near-expiry stock and reduces risk of dispensing errors.	
FR-002	Functional	The system shall allow the Pharmacy Clerk to record new batch details (batch number, quantity, expiry date, supplier) upon stock receipt.	High	Pass: A new batch record with all four fields is saved and visible in inventory within 5 seconds. Fail: Batch saved with a missing mandatory field.	Accurate batch data is the foundation for FEFO dispensing and expiry tracking.	
FR-003	Functional	The system shall automatically flag batches nearing expiry (within a configurable threshold, default 30 days) on an expiry alert dashboard.	Medium	Pass: All batches expiring within the threshold appear on the dashboard within 1 hour of crossing the threshold. Fail: A batch within threshold is absent from the dashboard.	Enables proactive clearance/return of stock before it expires, reducing financial loss.	
FR-004	Functional	The system shall automatically generate a purchase order request when an item's available stock quantity falls below its defined reorder threshold.	High	Pass: A draft purchase order is created within 10 minutes of stock dropping below threshold. Fail: Stock remains below threshold for over 1 hour with no purchase order generated.	Prevents stock-outs of critical medicines and automates a manual, error-prone task.	
FR-005	Functional	The system shall route every auto-generated purchase order to the Pharmacy Manager for approval before it is dispatched to the Inventory Supplier.	High	Pass: No purchase order reaches 'Dispatched' status without a recorded Manager approval action. Fail: An order is dispatched without an approval record.	Ensures financial and clinical accountability before committing to a supplier order.	
NFR-001	Nonfunctional (Performance & Security)	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the safety threshold, meeting defined latency and security benchmarks.	High	Pass: Benchmarking tests confirm target latency (<10 min generation time) and security standards (authenticated, encrypted transmission) under simulated peak load. Fail: Latency exceeds target or transmission is unauthenticated/unencrypted.	Guarantees timely, secure re-ordering even under high transaction volume.	
NFR-002	Nonfunctional (Security)	The system shall encrypt all supplier and inventory data at rest and in transit using industry-standard encryption (AES-256 at rest, TLS 1.2+ in transit).	Medium	Pass: Security audit confirms all stored and transmitted data is encrypted. Fail: Any unencrypted data store or channel is found.	Protects sensitive commercial and health-related inventory information from breach.	

Use-Case Flow

Use Case: UC-05 — Generate Purchase Order

Primary Actor: System (triggered), verified by Pharmacy Manager | Related Use Cases: «include» UC-07 Approve Purchase Order; extended by UC-06 Apply Emergency Reorder

Preconditions

- The Pharmacy Clerk has recorded at least one valid stock batch (UC-03) for the item in question.
- The item's reorder threshold has been configured in the system.
- The Inventory Supplier's catalog and contact details are active in the system.

Postconditions

- Success: A purchase order record exists with status 'Pending Approval' and is linked to the triggering low-stock item.
- Failure: No purchase order is created, and the low-stock condition remains flagged on the dashboard for manual follow-up.

Main Success Scenario

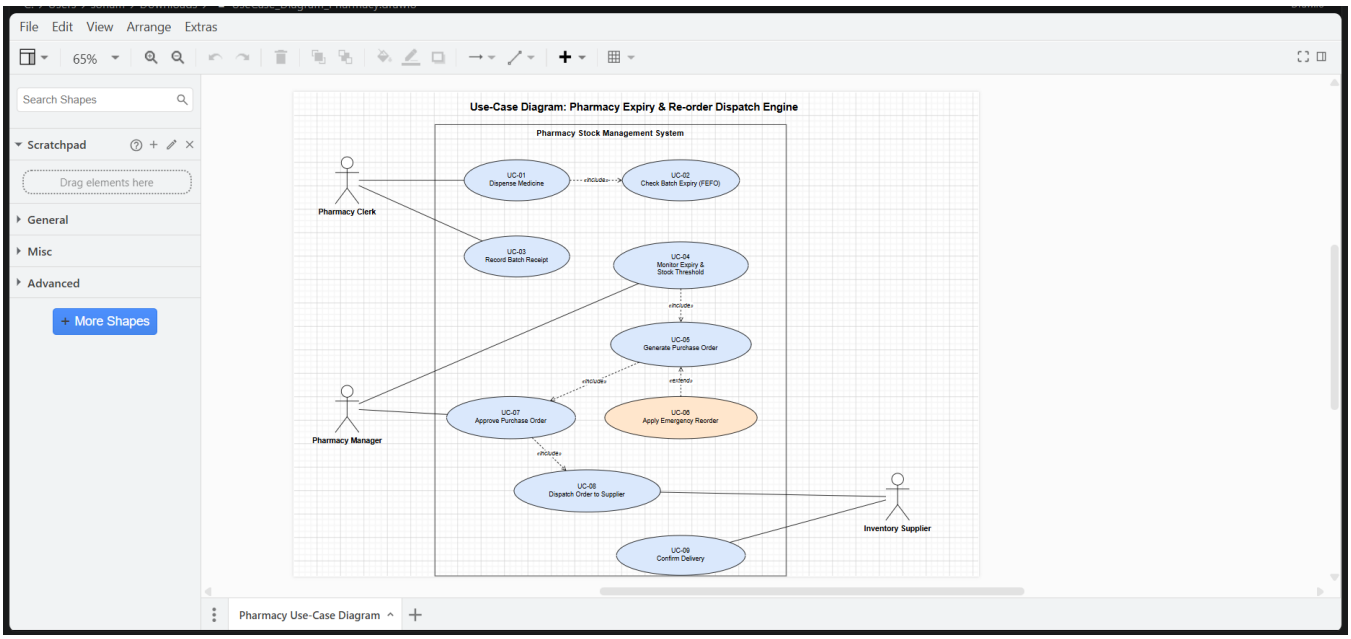
1. System (UC-04 Monitor Expiry & Stock Threshold) detects that an item's available quantity has fallen below its configured reorder threshold.
2. System retrieves the preferred Inventory Supplier and last-ordered unit price for the item.
3. System auto-populates a draft purchase order with item code, required quantity, supplier, and unit price.
4. System sets the purchase order status to "Pending Approval" and timestamps the creation.
5. System notifies the Pharmacy Manager of the pending purchase order (UC-07 Approve Purchase Order is triggered).
6. Pharmacy Manager reviews and approves the order.
7. System sets order status to "Approved" and proceeds to dispatch (UC-08).
8. Use case ends successfully.

Alternate Flow — 6a. Manager Rejects the Purchase Order

1. Pharmacy Manager reviews the draft order and rejects it (e.g., incorrect quantity or supplier).
2. System sets the order status to "Rejected" and records the Manager's rejection reason.
3. System returns the item to the expiry/stock alert dashboard as still requiring action.
4. Pharmacy Clerk or Manager may manually re-initiate a corrected purchase order.
5. Use case ends without a dispatched order.

Alternate Flow — 1a. Emergency Stock-Out (extension: UC-06 Apply Emergency Reorder)

1. If stock reaches zero (critical/emergency threshold) rather than the standard reorder threshold, UC-06 extends this use case.
2. System marks the purchase order as "Urgent" and shortens the Manager approval SLA.
3. Flow rejoins the main scenario at step 5 (notify Pharmacy Manager).


File Edit View Arrange Extras
65%
Search Shapes
Scratchpad
Drag elements here
General
Misc
Advanced
More Shapes
Use-Case Diagram: Pharmacy Expiry & Re-order Dispatch Engine
Pharmacy Stock Management System
Pharmacy Clerk
Pharmacy Manager
Inventory Supplier
UC-01 Dispense Medicine
UC-02 Check Batch Expiry (FEFO)
UC-03 Record Batch Receipt
UC-04 Monitor Expiry & Stock Threshold
UC-05 Generate Purchase Order
UC-06 Apply Emergency Reorder
UC-07 Approve Purchase Order
UC-08 Dispatch Order to Supplier
UC-09 Confirm Delivery
«include»
«extend»
Pharmacy Use-Case Diagram